

Harshith Reddy Nalla

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AI Engineer & Full Stack Developer experienced in building intelligent, scalable applications using PyTorch, FastAPI, React, and Spring Boot. I specialize in transforming complex deep-learning models into fast, production-ready systems. With hands-on experience in NSF-funded research, model deployment, and end-to-end system design, I focus on creating practical AI tools that improve decision-making and deliver real user impact.

EXPERIENCE

Research Assistant

06/2025 – Present

Computer Science Department. University of South Dakota

- Conduct AI research in medical diagnostics, developing deep-learning models (CNNs, Transformers, Mamba) for pancreatic cancer and dental imaging to improve early detection.
- Build and deploy full-stack medical AI systems using React, Spring Boot, FastAPI, Docker, and Hugging Face, enabling real-time inference, 3D visualization, and report generation.
- Improve model accuracy and reliability by enhancing data preprocessing, training pipelines, evaluation metrics, and visualization tools, making research results publication-ready.
- Contributed to benchmarking experiments, result analysis, and scientific writing for research papers, including accepted publications in medical imaging AI.

Student Government Association

08/2023 – 08/2024

Secretary of Finance Department. University of South Dakota

- Elected to the SGA senate through campus-wide voting.
- Work with other senators and the executive team to address student concerns and improve campus life.
- Represent student interests, share their views on key issues, and volunteer two hours each week for office hours.
- Help manage and decide how student funds are allocated to campus organizations and events.

Projects

Liver Profile AI - Automated Liver Segmentation & Analysis, developed an AI-based system using PyTorch, 3D CNNs, and Mamba architectures to perform automated liver segmentation and anomaly detection from CT/MRI scans. Integrated the model into a FastAPI backend and deployed it on Hugging Face Spaces with real-time visualization, enabling fast inference, Dice score evaluation, and interactive overlays for clinical interpretation.

UniVibe - Selected for the prestigious [NSF I-Corps program](#) as part of the UniVibe team, contributing as a Frontend Developer. This in-progress campus social networking platform, built with React and TypeScript, is designed to enhance student engagement across universities by helping students connect, share, and collaborate through a clean, modern, and responsive interface. Designed and built core UI features while conducting customer discovery to validate market needs and refine the business model.

PanInsight - A full-stack medical web application built with React, Spring Boot, Python, Pytorch, CNNs to detect pancreatic cancer from medical scans, featuring lesion overlays, confidence scores, downloadable reports, and an integrated AI chatbot for better understanding of results and next steps.

DentiMap - A full-stack web application that uses Python, PyTorch, and CNNs to analyze dental X-rays for quick, AI-driven diagnostics. Built with Java Spring Boot and a responsive React frontend, providing instant results with a clean, user-friendly interface.

SKILLS

- **Languages:** Java, Python, TypeScript, JavaScript, SQL
- **Frameworks & Libraries:** Spring, Spring Boot, Junit and Mockito, Pytorch, TensorFlow, Scikit-learn, React, Tailwind CSS, Hugging Face Transformers.
- **Tools:** Git & GitHub, Docker, Kubernetes, Hugging Face Spaces, AWS (AWS pipeline, ECS, EKS, EC2, S3).

PUBLICATIONS

When CNNs Outperform Transformers and Mambas: Revisiting Deep Architectures for Dental Caries Segmentation

Aashish Ghimire, Jun Zeng, Roshan Paudel, Nikhil Kumar Tomar, Deepak Ranjan Nayak, [Harshith Reddy Nalla](#), Vivek Jha, Glenda Reynolds, Debesh Jha

AAAI 2026 Workshop on AI4Research — arXiv: <https://arxiv.org/pdf/2511.14860>, Nov 2025

EDUCATION

University of South Dakota.

BS, Computer Science - AI

GPA: 3.565

01/2023 – 05/2027

[DSA with Java](#)

Professional Certificate

05/2025